

Application of Artificial Neural Network Model in Estimation of Wave Spectra

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Abstract: Estimation of a wave spectrum from specified values of significant wave heights and average zero cross periods is traditionally made through empirical equations like those of Pierson–Moskowitz and Jonswap. This technical note discusses an alternative scheme based on artificial neural network. Wave spectral distribution over various wave frequencies was obtained for given values of significant wave height and period using feedforward back-propagation network. The rider buoy data at a site off the United States coast monitored by the National Data Buoy Center was used as a basis for network development. Qualitative as well as quantitative comparisons of the network-yielded spectra with target spectra indicated that the developed network could model the wave spectral shapes in a better way than commonly used theoretical spectra.

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Introduction

The one-dimensional frequency spectrum of waves often forms a prerequisite to trials in structural design, simulation of random waves in a laboratory, as well as study of a variety of coastal processes, like wave refraction and sediment transport. The spectrum of waves can be derived from a specified design value of the significant wave height alone or in combination with that of the average wave period (normally made available to designers by analysis of visual observations, wind–wave relations, or statistical analysis of wave heights) by using empirical relationships, like those of Pierson–Moskowitz (PM), Jonswap, and Scotts. Many of such equations are available to the user, but the choice of one of them is often subjective. Investigations in the past have shown that many times these spectra fail to generalize the actual site conditions. In particular works done in India by Dattatri et al. (1977) and Narasimhan and Deo (1979) have shown inadequacy of the most common spectra of PM and Jonswap to Indian locations. The theoretical spectra had been derived on the basis of statistical curve fitting to field observations. In the recent past, soft computing techniques like artificial neural network (ANN) have proved to be a better alternative to many statistical schemes, e.g., Karunanithi et al. (1994) and Thirumalaiah and Deo (2000). This is presumably due to the ability of ANN to catch the hazy input–output dependency in a “model-free” and “data-oriented”

manner with considerable flexibility and adaptability.

The present work was directed toward derivation of the wave spectrum from only two overall wave parameters, namely, significant wave heights and average wave periods on the basis of ANN as well as checking the accuracy of such a network vis-à-vis traditional theoretical forms.

Acquisition of Data

The data were compiled from the U.S. National Data Buoy Center’s website (www.seaboard.ndbc.noaa.gov). They pertained to a location No. 42001, which is 180 NM south of southwest Pass, La. The choice of this site was not significant in this study and it was made considering the regularity or continuity in reported values. The data were collected by a 10-m discus buoy. The depth of water at the site was 3246 m. The hourly measurements of surface spectral density, $S(f)$, corresponding to different frequencies, f , ranging from 0.1 to 0.425 Hz, resolved at an interval of 0.01 Hz, along with values of the significant wave height, H_s , and the average time period, T_z , calculated on the basis of standard deviation of surface displacements, were downloaded for a period of 26 months from January 2002 to February 2004 in isolated events. In total 721 time histories randomly but evenly distributed over the temporal domain were considered.

Division of the total samples into network training and testing sets is generally done on an empirical basis and in the present study 85% of all measurements were used for training and the remaining ones were set aside for testing purposes. The number of training pairs was 612 and that of testing pairs was 109. The choice of training and testing data had been made randomly over the available sample size, but taking care that a variety of spectral shapes and intensities were involved in both sets.

Development of Networks

An ANN involves a number of simple processing elements (neurons or nodes) arranged in different layers, and connected

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together to form a network. The neurons in a given layer sum up their inputs, effect a nonlinear data squashing process (through say the sigmoid function), and transmit a single output to all processing elements in the next layer via the connections. One of the most common types of ANNs is the multilayered perceptron (MLP), which consists of at least three layers of neurons namely, input, hidden, and output, and, where the information flows in the forward direction. Such a feedforward network can be conveniently trained using a back-propagation algorithm, which in its simplest form, involves minimization of the error between realized and target outputs according to the steepest or gradient descent approach. [Refer to textbooks like Wu (1994), Wasserman (1998), and Kosko (1992) for details.] In such an approach the correction to a connection weight, Δw , is made proportional to the rate of change of global error, E , with respect to that weight, w , i.e.

$$\Delta w \propto (\partial E / \partial w) \quad (1)$$

The input to the present network was in the form of significant wave height and average zero-cross wave period, characterizing the short-term sea state in an average manner, while the output from the network belonged to values of the spectral density, $S(f)$, at a large number of specified wave frequencies. The number of input nodes was thus two. The number of output nodes was determined considering a tradeoff between preservation of details of the spectrum on one side and the large size of the network on the other. This value was thus kept as 30. The general MLP network was trained using the ordinary back-propagation and in some of its variants (Demuth et al., 1998) in order to make sure that adequate training was imparted for such network architecture. The neural network toolbox under MATLAB environment (Math-Works Inc. 1999) was used as the main processor.

The number of hidden layers as well as the number of neurons in the general MLP network was calculated by trials aimed at getting the minimum mean square error at the end of a sufficiently large number of iterations. The resulting number of hidden layers was thus two, with the first layer of three neurons and the second layer of six neurons each.

Network Validation

After training the network, resultant weights were saved. Using these weights testing of the network was carried out in order to assess its predictive capability for unseen inputs. This was achieved by means of a qualitative comparison (through spectral plots) between the network-yielded and the "true" or target spectral densities directly calculated from the sea surface elevation

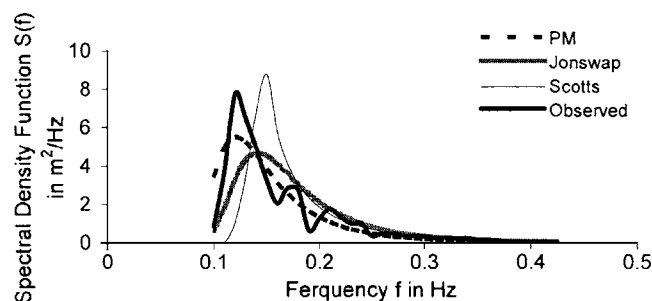


Fig. 1. Theoretical and observed spectra (1800 hrs on November 16, 2002)

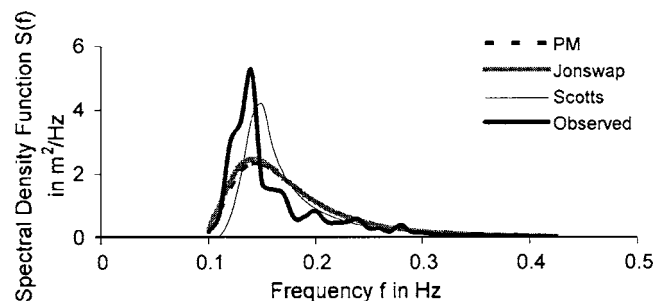


Fig. 2. Theoretical and observed spectra (1600 hrs on March 31, 2003)

records and also on the basis of evaluation of quantitative measures of correlation coefficient (CC), mean square error (MSE), and mean absolute error (MAE). (Please see Appendix I for relevant expressions.)

Results

Estimation Based on Theoretical Spectra

Before considering application of the networks it was thought worthy to examine how well existing empirical spectra represent the observed spectral shapes. The theoretical spectra considered were PM, Jonswap, and Scotts whose expressions as used in this study are given in Appendix II. These were worked out corresponding to given values of H_s , T_z . The value of spectral peak frequency ω_0 involved in the Scotts spectrum was first taken as an inverse of the average zero cross-wave period. However this gave a high level of mismatch with observed spectra. In order to bring the Scotts spectra close to the observed ones the ω_0 value obtained in this way was multiplied by an empirical factor of 1.2. Similarly the area under the Jonswap spectrum obtained with the help of equations given in Appendix II was much higher than that under the true spectrum. In order to remove this anomaly an empirical correction was made to all ordinates of the Jonswap spectra. It was found that while theoretical expressions used without any empirical corrections produced highly unacceptable spectral shapes, the corrected spectra, shown in the form of two examples as in Figs. 1 and 2, also exhibited large variations from the true spectra.

Application of Network

Figs. 3 and 4 show two typical examples of how the network-yielded spectra compared with the actual observed ones. It is clear

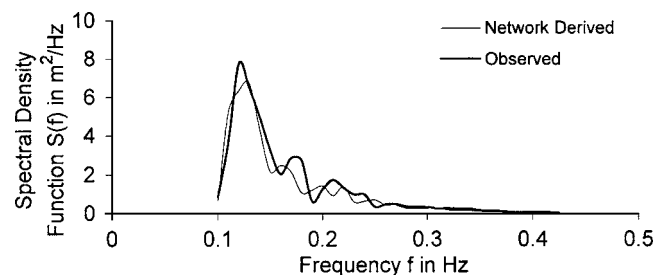


Fig. 3. Network-derived and observed spectra (1800 hrs on November 16, 2002)

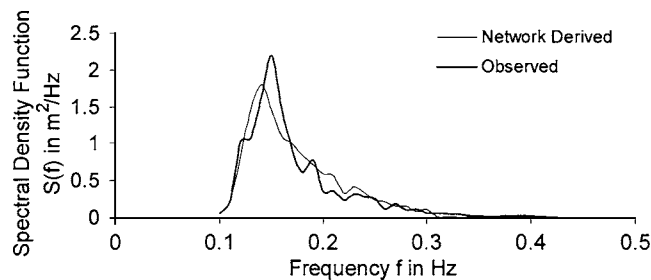


Fig. 4. Network derived and observed spectra (2200 hrs on March 31, 2003)

that the ANN was able to satisfactorily predict the shape of the wave spectrum from the short-term average parameters of H_s and T_z . The network predicted the ups and downs of the true spectral curves in a fairly close manner. However the exact peak density was found to be generally underestimated, like most theoretical spectra. The networks could be seen to be relatively successful in averaging out the important high frequency portion of the true spectrum.

It is well known that the multi peaked spectra are difficult to model by conventional theoretical spectral equations. In the case of current application of the ANN (see the example of Fig. 5) it was found that the predicted spectra fairly averaged out the multiple peaks and further that they fairly approximated the high-frequency portion. However it appeared that if the observed spectral fluctuations are to be properly accounted for, additional parameters other than H_s and T_z might be required in multi peaked spectral predictions.

The above qualitative comparison was further quantitatively confirmed by the high values of CC between the true and network-yielded spectral densities and further by low values of the MSE and MAE. The CC across the entire testing set stationed around 0.91 while the MSE and the MAE revolved around the values of 0.12 and 0.14. All these measures thus suggest a satisfactory performance of the ANN.

As mentioned earlier, the peak spectral density was found to be generally underestimated, like most theoretical spectra. This problem was overcome to some extent by considering a single empirical multiplying constant obtained after trial and error. The resulting constant value of 1.107 was universally used to multiply the network-derived peak density and the two nearby spectral density functions of the spectrum, during the training process, keeping the rest of the densities undisturbed. This considerably helped in peak predictions as exemplified in Fig. 6.

The development of the network as above was based on training imparted with the help of actual observations of the wave time

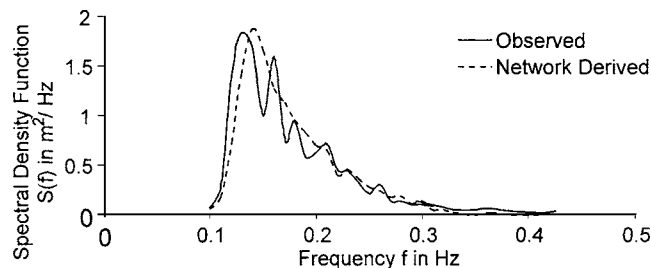


Fig. 5. Network derived and observed spectra (0800 hrs on December 12, 2004)

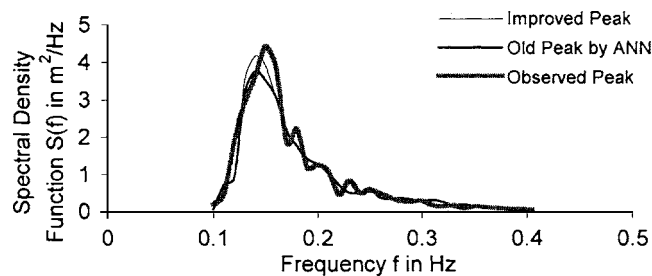


Fig. 6. Network derived and observed spectra after peak improvement (0300 hrs on February 8, 2004)

history. At many places over the world and for most new projects such data collection may not be possible. In these situations the network could be trained with the help of the theoretical spectra, which can be derived from the H_s , T_z values, which in turn should be available from visual or wind sources. Further, in such cases, average (spectral densities) of more than one such spectrum instead of a single form of them could be used to better advantage because a single spectrum may not be representative of the true conditions for all the times and at all the locations.

For every training spectrum involved in the preceding analysis the spectral density averages of the PM, Jonswap, and Scotts were worked out and the training was redone with these density values. The trained network was thereafter tested for the unseen input. Fig. 7 shows the results as an example. The CC for the entire testing set varied from 0.80 to 0.86 while the MSE changed from 0.22 to 0.41 and the MAE varied from 0.24 to 0.32. Although the match between the target and network-yielded spectra was not as good as the one corresponding to the earlier case of training based on "true" spectra, it was noticed that the network can learn from the theoretical spectra as well, albeit with reduction in the accuracy of results.

In order to see if better results were possible by adoption of some innovative alternatives the network was trained by: (1) adding a few variants of the ordinary back-propagation learning schemes mentioned in Demuth et al. (1998); (2) introducing some level of stochastic perturbations in the input; (3) training one value of the spectral density at a time; and (4) using other network architectures. However the improvements noted could not be generalized for all the test cases and hence use of these alternatives cannot be recommended before more detailed experimentation is carried out.

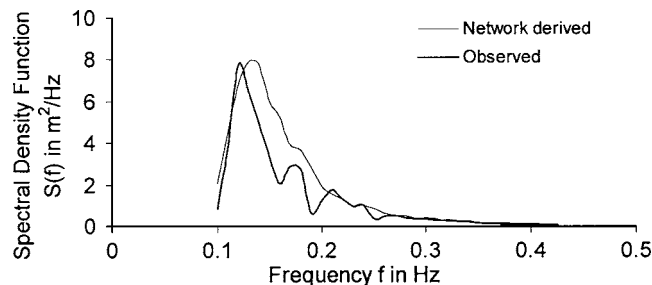


Fig. 7. Network-derived and observed spectra trained by averaged theoretical spectra (1800 hrs on November 16, 2002)

Conclusions

The foregoing sections presented development of an ANN in order to estimate the wave surface spectral density function over a wide range of wave frequencies out of the specified values of the significant wave height and the average zero-cross wave period, which are normally made available to designers by visual observations, wind-wave relations, or statistical analysis of wave heights. The trained network when tested for unseen inputs showed that the ANN can be a viable option in order to estimate the shape of wave spectrum. This was evident from the high values of the coefficients of correlation and low values of the mean square as well as mean absolute errors between the network-predicted and the target spectral density.

The comparison with observed values revealed that the network-predicted spectral shapes were more satisfactory than those yielded by the theoretical spectra of PM, Jonswap, and Scotts.

While use of the available wave time history could be more beneficial for training, the network can also reasonably learn from the theoretical spectra, albeit with reduction in resulting accuracy.

Appendix I. Error Measures Used

The expressions of the error measures used are as below

$$CC = \frac{\sum x * y}{\sqrt{\sum x^2 * \sum y^2}} \quad (2)$$

where $x = X - \bar{X}$, $y = Y - \bar{Y}$; X =observed values; $\bar{X}$ =mean of X ; Y =predicted value; and $\bar{Y}$ =mean of Y . The summation in the above equation as well as in the following two equations is carried out over all n number of testing patterns:

$$MSE = \frac{\sum (X - Y)^2}{n} \quad (3)$$

$$MAE = \frac{\sum |X - Y|}{n} \quad (4)$$

Appendix II. Theoretical Spectra Used

The Pierson-Moskowitz spectrum model is as below

$$S(\omega) = \alpha g^2 \omega^{-5} \exp \left[-0.74 \left(\frac{\omega U_w}{g} \right)^{-4} \right] \quad (5)$$

where $S(\omega)$ =spectral density function for circular wave frequency ω ; $\alpha=0.0081$; g =acceleration due to gravity; U_w =wind speed, related to H_s through the equation

$$U_w = 6.8471 \sqrt{H_s} \quad (6)$$

The Jonswap spectrum, as given in modified form by Goda (1979) and reported in Chakrabarti (1987) is as below

$$S(\omega) = \alpha^* H_s^2 \frac{\omega^{-5}}{\omega_0^{-4}} \exp \left[-1.25 \left(\frac{\omega}{\omega_0} \right)^{-4} \right] \gamma \exp [-(\omega - \omega_0)^2 (2\tau^2 \omega_0^2)] \quad (7)$$

where ω =peak frequency; γ =peak enhancement factor calculated from given H_s and T_z as per Chakrabarti (1987); and τ =shape factor=0.07 for $\omega \leq \omega_0$ and =0.09 for $\omega > \omega_0$

$$\alpha^* = \frac{0.0624}{0.230 + 0.0336\gamma - 0.185(1.9 + \gamma)^{-1}} \quad (8)$$

The Scotts spectrum is given by

$$S(\omega) = \left\{ 0.214 H_s^2 \left[-\frac{(\omega - \omega_0)^2}{0.065(\omega - \omega_0 + 0.26)} \right]^{1/2} \right. \\ \left. \text{for } -0.26 < (\omega - \omega_0) < 1.65 \right. \quad (9)$$

where ω_0 =peak spectral frequency obtained by empirically relating it to T_z .

References

- Chakrabarti, S. K. (1987). *Hydrodynamics of offshore structures*, Springer, Berlin.
- Dattatri, J., Raman, H., and Shankar, N. (1977). "Comparison of Scott spectra with ocean wave spectra," *J. Waterw., Port, Coastal, Ocean Div., Am. Soc. Civ. Eng.*, 103(3), 375–378.
- Demuth, H., Beale, M., and Hagen, M. (1998). *Neural networks toolbox—user's guide*, The MathWorks Inc., Natick, Mass.
- Goda, Y. (1979). "A review on statistical interpretation of wave data." *Rep. of the Port and Harbor Research Institute No. 18(1)*, Japan.
- Karunanithi, N., Grenney, W. J., Whitley, W. J., and Bovee, K. (1994). "Neural networks for river flow prediction." *J. Comput. Civ. Eng.*, 8(2), 201–220.
- Kosko, B. (1992). *Neural networks and fuzzy systems*, Prentice-Hall, Englewood Cliffs, N.J.
- MathWorks Inc. (1999). *MatLab the language of technical computing*, Version 6, Natick, Mass.
- Narasimhan, S., and Deo, M. C. (1979). "Spectral analysis of ocean waves—A study." *Proc., 4th Conf. Civil Engineering in the Oceans IV*, ASCE, New York, 877–892.
- Thirumalaiah, K., and Deo, M. C. (2000). "Hydrological forecasting using neural networks." *J. Hydrologic Eng.*, 5(2), 180–189.
- Wasserman, P. D. (1998). *Neural computing—theory and practice*, Van Nostrand Reinhold, New York.
- Wu, J. K. (1994). *Neural networks and simulation methods*, Marcel Dekker, New York.